LEVEL UP ACADEMY

Class 9 | CBSE / GSEB | Science

Unit Test: Mixtures, Solutions and Separation

Chapter / Topic: Is Matter Around Us Pure (topic-based)

Name: _____ Roll No.: _____ Date: _____

Time: 50 minutes **Max. Marks: 25**

General instructions:

1. All questions are compulsory. Internal choice is given in some questions.
2. Section A: 1 mark each. Section B: 2 marks each. Section C: 3 marks each. Section D: 5 marks. Section E: case-based, 4 marks.

SECTION A

6 marks

Choose the correct option.

Q1. Which of the following is a homogeneous mixture? **[1]**

(a) Sand in water (b) Oil in water
(c) Salt solution (d) Chalk in water

Q2. Which of these shows the Tyndall effect? **[1]**

(a) Salt solution (b) Sugar solution
(c) Copper sulphate solution (d) Milk

Q3. Camphor can be separated from common salt by: **[1]**

(a) filtration (b) sublimation
(c) distillation (d) chromatography

Q4. Which of these is a chemical change? **[1]**

(a) Rusting of iron (b) Melting of ice
(c) Dissolving sugar in water (d) Boiling of water

Q5. Brass is: **[1]**

(a) an element (b) a homogeneous mixture (alloy)
(c) a compound (d) a heterogeneous mixture

Q6. **Assertion (A):** A sugar solution does not show the Tyndall effect.
Reason (R): The particles in a true solution are too small to scatter light. **[1]**

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

SECTION B

4 marks

Q7. Classify each as an element, a compound or a mixture: water, air, copper, sugar. **[2]**

OR

What is a saturated solution? How does the solubility of most solids change with temperature?

Q8. Find the mass of glucose in 250 g of a 20% (mass by mass) glucose solution. **[2]**

SECTION C

6 marks

Q9. Suggest a method to separate: **[3]**

(i) butter from curd. **[1]**

(ii) iron filings from sulphur powder. **[1]**

(iii) salt from sea water. **[1]**

Q10. What is chromatography? Explain how it separates the colours in ink, and give two uses. **[3]**

OR

How would you separate a mixture of acetone and water? Name the method and explain why it works.

SECTION D

5 marks

Q11. Answer the following: **[5]**

(i) Compare true solutions, colloids and suspensions on particle size, visibility of particles, and stability. **[3]**

(ii) Classify as a true solution, colloid or suspension: milk, chalk in water, salt solution, smoke. **[2]**

SECTION E

4 marks

Read the passage and answer the questions that follow.

Q12. While making tea, Priya boiled water, added tea leaves, sugar and milk, and then poured the tea through a strainer into a cup. The tea leaves were left behind in the strainer. **[4]**

(i) In dissolving the sugar, which substance is the solvent? **[1]**

(ii) What is the residue and what is the filtrate in this process? **[1]**

(iii) Name the separation method used, and say whether sugar dissolving in water is a physical or a chemical change. **[2]**

~ All the best! ~